



## 1U *MICRO* Compact System

Installation Manual  
V1.3

Manufactured by Enatel Ltd.  
321 Tuam Street  
PO Box 22-333  
Christchurch  
New Zealand

Phone +64-3-366-4550  
Fax +64-3-366-0884  
Email [sales@enatel.net](mailto:sales@enatel.net)  
[www.enatel.net](http://www.enatel.net)

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## Receiving Instructions

### **CAUTION:**



For your protection, the following information and the product manual should be read and thoroughly understood before unpacking, installing and using the equipment.

We present all equipment to the delivering carrier securely packed and in perfect condition. Upon acceptance of the package from us, the delivering carrier assumes responsibility for its safe arrival to you. Once you receive the equipment, it is your responsibility to document any damage the carrier may have inflicted, and to file your claim promptly and accurately.

### **Package Inspection**

- Examine the shipping crate or carton for any visible damage: punctures, dents and any other signs of possible internal damage.
- Describe any damage or shortage on the receiving documents and have the carrier sign their full name.

### **Equipment Inspection**

- Within fifteen days, open crate or carton and inspect the contents for damages. While unpacking, be careful not to discard any equipment, parts or manuals. If any damage is detected, call the delivering carrier to determine the appropriate action. They may require an inspection.  
*Save all the shipping materials for the inspector to see!*
- After the inspection has been made and you have found damage, call us. We will determine if the equipment should be returned to our plant for repair or if some other method would be more expeditious. If it is determined that the equipment should be returned to us, ask the delivering carrier to send the packages back at the delivering carrier's expense.
- If repair is necessary, we will invoice you for the repair so that you may submit the bill to the delivering carrier with your claim forms.
- It is your responsibility to file a claim with the delivering carrier. Failure to properly file a claim for shipping damages may void warranty service for any physical damages later reported for repair.

### **Handling**

Handle the equipment with care. Do not drop or lean on front panel or connectors. Keep away from moisture.

### **Identification Labels**

Model numbers are clearly marked on all equipment. Please refer to these numbers in all correspondence with Enatel.

## 1.0 SCOPE

This manual covers essential information for the installation and commissioning of the 1U *MICRO* Compact Enatel Compact DC Power System Range (see Appendix for individual model specifications).

System set-up for the rectifiers, alarms etc., are provided in separate manuals for the SM35/6 supervisory module and RM848 rectifier.

All installation and maintenance must be carried out by suitably qualified personnel.

**Note:** The 1U *MICRO* Compact system is available with positive earthing (fixed) or positive/negative earthing (user selectable - negative earth as default). The installation manual covers both earthing varieties, the standard system is assumed to be fixed positive earthing. Where parameters and settings differ between systems, the negative earthed system parameters will be specified within parenthesis i.e.( ).

## 2.0 SYSTEM OVERVIEW

Systems with a model number beginning with PSX24 can hold three rectifier modules and have a maximum power output of 2.4kW, producing a maximum current output of 44.4A at 54Vdc.



The system is intended to be a complete power system in a box, so no connections need to be made internally. AC connection is via rear exiting lead with all the DC (Load and Battery) connections are made at the front of the unit. Alarm connections are accessible from the front.

The system is designed to be extremely simple to install and set up.

The following is a summary of the system:

- Overall size is 483mm wide (19" standard mounting) x 44.5mm high (1U) x 280mm deep (within ETSI specification)
- Optional ETSI mounting tabs included
- Up to 3 RM848 series rectifiers - may be packaged separately
- SM3x supervisory module (fully integrated in the system)
- Battery Low Voltage Disconnect fitted as standard (80A rating).
- Up to 2\*x 30A Battery Circuit Breaker, this may be specified as different values (from 2A to 30A) at time of order.
- Up to 4\*x 10A Load Circuit Breaker, this may be specified as different values (from 2A to 30A) at time of order.
- System weight is approximately 3.6kg without rectifiers, and 5.6kg with three rectifiers fitted.
- Single phase lead supplied for input termination (phase, neutral and earth)
- Front or rear cable access

\*values shown are maximum values and depend upon model selected. Please see model numbers and descriptions in the appendix of this manual.

**Note:** This system is supplied with the AC and DC earths connected. The standard system (+ve earth system) output has the DC Common (earth connection) in the positive side of the circuit.

The user selectable, positive/negative earth system has the DC Common (earth connection) in the negative side of the circuit, but can be changed by following the instructions in the DC earthing section of this manual.

The earth link can be completely removed from the system to isolate earths. Please see the DC earthing section of the manual.

## 3.0 INSTALLATION

### 3.1 Unpacking & Installing in 19" Rack

Upon unpacking, check that the unit is not damaged, and that you have the required number of rectifiers.

Remove the transport bracket supplied attached to the top of the system.

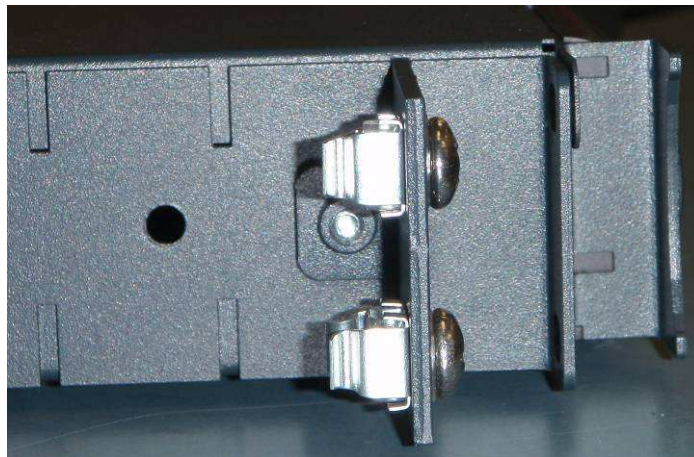
The unit fits into a standard 19" mounting frame. The mounting screws should be M6, however M5 may be used with washers. Be sure to mount the unit in the 19" frame squarely if M5 screws are used.

To fit the cover, remove the screws securing the top of the system and place on, allowing the rear tabs to lock into the slots provided. Secure the cover to the chassis by replacing the screws. The rack will be able to securely hold the 1U system with just the bottom two screws fitted. Brace cabinet mounting if necessary.

Please note the complete system weight is 5.6Kg. Ensure the 19" mounting rails are able to withstand mounting of the system.

### 3.2 Installing in ETSI Rack

Fit ETSI mounting tabs to each side of the compact system chassis. Attach the tabs to the desired depth with the included M5 Screws. Screws are accessible from the inside of the system and as such ETSI mounting tabs can only be fitted prior to installing in a rack.



*Fig 1, Fitting of ETSI rack mount tabs*



*Fig 2, Fitting of ETSI rack mount tabs*

**Note:** Fitment of tabs in some positions will require the temporary removal of the DC Battery connectors and SM3x monitor. These can be replaced once fitment is complete.

### 3.3 AC Cabling

The AC cables are clearly labelled at the rear of the system (see Fig 3). The AC cable provided is 2.5mm<sup>2</sup> cables and suitable terminals should be used.

AC connection internally is via 6.4mm QC tabs. Longer cables can be easily fitted by replacing the existing cable. Ensure that the following AC colour coding is maintained:

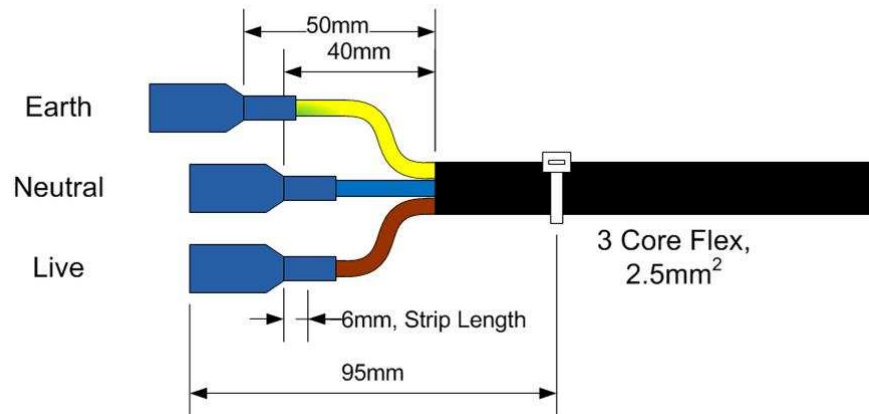


Fig 3, AC Cable Detail

Connections should allow for a maximum single phase AC supply of 15A (@ 175VAC) (see next section).

The AC earth is internally bonded to the system chassis.

DC Common is connected to the AC earth by a PCB placement, depending on required earthing connection. Please see the DC earthing section of this manual for more detail.

#### 3.3.1 Upstream Over-current Protection

There are two considerations to take into account when selecting an appropriate fuse/circuit breaker.

- Firstly, the upstream protection should protect the downstream cable from overload situations and
- Secondly, discrimination should be maintained with the downstream device fuses.

##### (i) Cable Rating

The maximum current drawn by this DC power system is 15A (5A per rectifier at a minimum input voltage of 175V<sub>ac</sub> and full output power). The upstream protection device must be able to supply this load under all conditions without tripping. Therefore, typically at least 20% headroom is allowed for in the protection device, making its minimum rating 18A.

**Note:** The current carrying capacity of cables is dependent on the type of cable used. Please check with your local supplier and local regulations for appropriate sizing. For convenience, the system is supplied with a 1.5m long, 3-core, 2.5mm<sup>2</sup> flex already attached. This has a current carrying capacity of 20A.

##### (ii) Discrimination

Discrimination ensures that the upstream circuit breaker or fuse does not blow if a rectifier input is faulty (in a short circuit manner). Therefore it is important to ensure the upstream protection discriminates with the internal fuse of the rectifier. The fuse used in the RM848 is a slow-blow 10A fuse. The tripping curve for this is shown in Appendix 2 at the rear of this manual.

A minimum circuit breaker to use for this system is a 20A, D-curve (note, a 20A C-curve breaker will **not** discriminate with the rectifier fuse). Therefore, when used with the 2.5mm<sup>2</sup> cable supplied, a 20A, D-curve breaker should be used.

Alternatively, a 32A C-curve breaker, or greater, can be used. However, AC cable provided may have to be replaced for a larger cable<sup>1</sup>.

If a fuse is used upstream, then any BS88 or NH g style fuse, of 20A or greater rating will discriminate.

### 3.4 DC Cabling

**CAUTION:** Use extreme care when fitting batteries & their connections. Remove all jewellery and rings from oneself prior to commencing the installation. Always use insulated tools when fitting batteries and take extreme care not to short terminals when working on them.



All DC Load cables terminate to the connectors at the front of the unit as shown in Figure 5. These terminals are all 6mm<sup>2</sup>. Load connections are of a push in design and cables can be simply pushed into the connector for termination after stripping to the appropriate length.

Removal requires a small screwdriver to be inserted into the rectangular hole above the connector whilst pulling on the cable.



Connector designators are available on the pull-out guide. See Figures 4 and 5 for detail.



Fig 4 & 5, DC Connector detail and pullout guide

The battery cabling connects through the SB50 style power-pole connectors. From here it goes directly to the appropriate circuit breaker, then via a Low Voltage Disconnect relay and current shunt to the internal live bus. This can be seen in the appropriate wiring diagram as the rear of this manual.

### 3.5 Alarm/Ancillary Cabling

Alarm and communication cables terminate directly into the connectors of the Supervisory Module, SM35 or SM36, which terminals are assessable by pulling the monitor forward to expose connections (see Fig 6 & 7).

Cables can be routed through the front of the system via the cable exit indicated. When routing the cables, ensure they are kept away from the AC and DC power cables when possible.

<sup>1</sup> NOTE: A larger breaker may be used even though in theory it may appear that the 2.5mm<sup>2</sup> wire is not fully protected. In fact it is protected on two accounts. Firstly it is protected by the rectifier input fuse (which is only a short distance away). Secondly, the rectifiers are power-limited on their input. Therefore, they can never be overloaded. As a result, the wire can never be over-loaded by the rectifier – it can only see fault current. As a result, depending on local authorities, only fault current protection may be catered for by the upstream protective device.



*Fig 6. For removal, pull monitor forward to release the ball catch.*



*Fig 7. SM3x cable access*

Relays 1- 6 can be used for normally open or normally closed states by jumper selection. The relay states labelled NO or NC are for their de-energised state. If an alarm is programmed for the relay to be normally energised (as may be possible in the case of a low voltage alarm where loss of power will put the alarm into its active state), then be sure to connect the remote wiring appropriately.

For full monitor functionality and operation information, refer to the appropriate monitor manual.

Uncoil the battery temperature sensor and place in the middle of the middle battery string. If the lead is not long enough, ordinary 2-core copper (approx.  $0.75\text{mm}^2$ ) wire can be used as an extension. The purpose of the battery temperature sensor is to monitor the ambient temperature of the batteries over long periods of time and adjust the rectifier output (float) voltage accordingly. As a result, it is not necessary to have the temperature sensor touching the batteries. If the Battery Temperature Sensor is removed a “battery temp fault” alarm is generated.

For remote communications and direct computer connection to the Compact System, refer to the monitor manual. These connections can be made via the mini-USB port on the front panel of the monitor (computer connection), and the Ethernet port (SM36 web-based communications only).

## 4.0 ALARM MAPPING TO VOLTS-FREE RELAYS

Relay 1 is pre-configured as the “Monitor Fail” alarm. This alarm activates if the monitor has a hardware fault or if software becomes corrupted.

All other relays can be mapped to different alarm conditions. The monitor manual details how these may be changed. On the standard Compact Systems alarms are preconfigured as follows:

Relay 2: Summary Non-urgent alarm

Relay 3: Summary Urgent alarm

Relay 4: User Configurable

Relay 5: User Configurable

Relay 6: User Configurable

As mentioned, if these mappings are not appropriate, they can be changed in the field to suit customer requirements.

## 5.0 CIRCUIT BREAKER FAIL MONITORING

The load circuit breakers are monitored electronically through an internal general purpose input on the monitor. The digital input will trigger an alarm when it is pulled to the system common (+ve or –ve depending on configuration) rail. This means that to operate the load must be connected. In this way, false alarms are avoided when no load is connected and all load circuit breakers are in the “off” position.

**Note:** This means that a residual voltage will be measured at the load terminal even when the circuit breaker is turned off. This is high impedance and does not present a hazard to the user.

The battery circuit breakers however, use voltage sense to detect tripping or whether they are turned off. This is because when a battery breaker is tripped, there may be very little voltage difference, making electronic fail detection problematic. Hence, if no battery is connected, the breaker must be “on” to clear the Battery Breaker Fail alarm.

## 6.0 LVD OPERATION

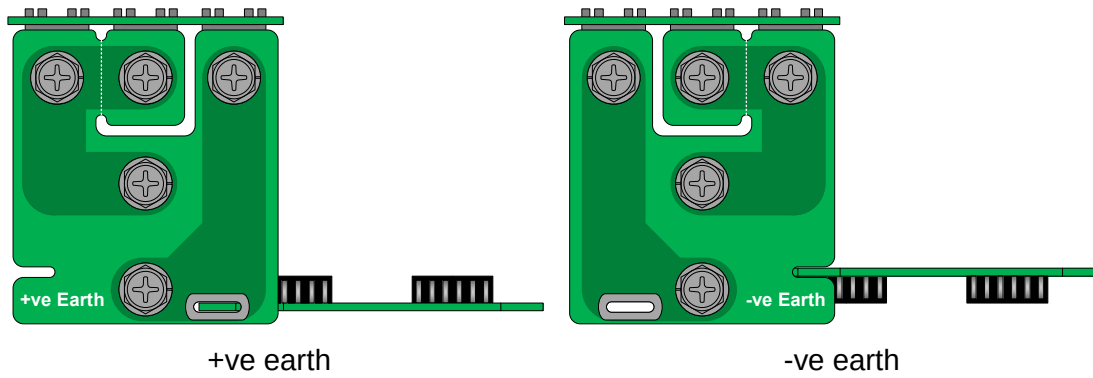
This system is configured with a single Low Voltage Disconnect contactor in the battery side of the circuit (see wiring diagram at rear of this manual).

The supervisory monitor unit is powered from both the rectifier side of the LVD contactor and battery source. Therefore when the low voltage threshold is reached and the LVD disconnects the battery, the SM3x will lose voltage sense (as voltage sense is measuring rectifier bus voltage) but still maintains operation for monitoring system. The LVD contactor will not re-engage until the rectifier power is restored (i.e., until the DC bus voltage is restored).

## 7.0 DC EARTHING

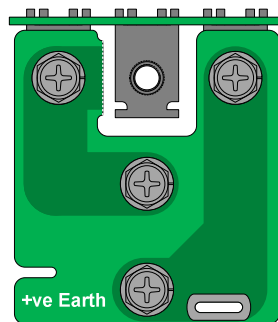
The 1U micro system is available with two DC Earthing options. The first is fixed by hardware as a positive earth system where the positive DC output is tied to AC earth. This cannot be changed in the field.

With the second option, AC earth can be tied to either the positive or negative output by simply inverting the +/- earth PCB. The monitor signals are modified by rotating the second PCB as shown in figures 8 & 9. The two PCB's mechanically interlock, preventing incompatible connections to be made.



*Fig 8 & 9, +ve earth and -ve earth options*

For systems that require no connection to earth, the PCB can be cut as indicated in figure 10 to isolate DC output from AC earth. This modification should be made with both +/- earth PCB's in the +ve earth position (or with the fixed positive earth system)



*Fig 10, isolated earth option*

## 8.0 COMMISSIONING

Use the following set of instructions as a guide, unless different procedures are recommended by local authorities.

### 8.1 System Pre-check:

1. System installation is completed.
2. Battery and load circuit breakers are turned off, and upstream AC has been turned off (i.e., system is completely de-powered).
3. Rectifier modules have been placed into their shelf positions, and pulled forward enough to disconnect them from the system.
4. Check that the protective AC Earth is connected as per local regulations.

**Note:** This system is available in either positive earthing (-48VDC) or negative earthing (+48VDC) configurations. Before connecting batteries or rectifiers ensure that the correct system has been specified and earthing is correct for your application.

5. Turn on AC upstream and check that voltage from phase to neutral are as expected.

### 8.2 Rectifier Start-up

1. Turn the upstream AC circuits on.
2. Fully insert first rectifier, wait for the rectifier to start and its power on LED to remain green.
3. Check the SM3x powers up, and indicates the system default float voltage on its display. If the audible alarm activates, press any button to silence it.
4. If a different system float voltage has been specified, set this at this time using the procedure specified in the SM3x Manual (either from the front panel or connected computer).
5. Fully insert the rest of the rectifiers ensuring they power up with only their green “power on” LED illuminated.
6. Check that the load and battery currents on the SM3x are 0 amps (+/- 1 or 2 amps).
7. Check that all SM3x configuration settings are correct (as per customer specification) with respect to:
  - Voltage levels
  - Alarm settings
  - Alarm mappings to the volts-free relays(refer to the SM3x manual for information on how to check these via the front panel or locally connected computer).

### 8.3 Battery Start-up

**Note:** It is important that battery circuit breaker connections should be made when the rectifiers are turned on and the system is “live”. This is because the system voltage and battery voltages will be similar, thus minimising any arcing during connection. This also prevents high current arcing due to the charging of the rectifier output capacitors.

1. Power up only one rectifier initially (to limit any damage if any connections are incorrect).
2. Measure the voltage across each battery string at the terminals of the Compact System. Ensure that the reading from the DC Common bus to the Battery Live Terminals is -48V (or +48V as per system specification).
3. Turn on each battery circuit breaker in succession while measuring the battery voltage and ensure that the voltage increases slightly to the system Float Voltage (typically the voltage will increase from 2-3V below float to float voltage. At this point the batteries will be drawing some current to bring them to a full state of charge.

### 8.4 Load Start-up

1. Ensure downstream load connections have been made and there are no loose/floating cables.

2. Turn on load circuit breaker, ensuring that the downstream equipment is being powered up as expected.
3. Ensure the system float voltage on the SM3x is at the level previously noted.
4. Ensure the load current is at a level expected (could be zero if loads downstream have not been connected).

**Note:** Prior to leaving the system after it has been commissioned, ensure all AC, DC *and* battery circuits are off. If it is required that the system is to be left on (to power load equipment, ensure rectifiers are left in their powered up state, and batteries are in circuit. This will prevent anyone leaving the batteries only powering the load (in which case the batteries would go flat).

## 8.5 DC System Commissioning Check-List

| DC SYSTEM COMMISSIONING CHECK-LIST |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |             |       |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------|-------|
| Site Name: _____                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      | Date: _____ |       |
| Tests Without Batteries Connected  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Measured/<br>Setting | Results     |       |
|                                    | Check Float Voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Meter: _____         | _____ V     | ✓ / ✗ |
|                                    | Check Load Current                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Meter: _____         | _____ A     | ✓ / ✗ |
|                                    | <b>Alarms:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                      |             |       |
|                                    | <p>Voltage thresholds can either be checked using an external power supply, or by adjusting the SM3x float voltage 0.1V above (or below for the low voltage alarms). It is recommended to have the batteries <b>disconnected</b>.</p> <ol style="list-style-type: none"> <li>1. Adjust the supply/float voltage to 55.7V &amp; observe the "High Float" alarm.</li> <li>2. Adjust the supply/float voltage to 57.7V &amp; observe the "High Load" alarm.</li> <li>3. Adjust the supply/float voltage to 52.7V &amp; observe the "Low Float" alarm.</li> <li>4. Adjust the supply/float voltage to 46.9V &amp; observe the "Low Load" alarm.</li> </ol> |                      |             |       |
|                                    | High Load Volts (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 57.6V                | _____ V     | ✓ / ✗ |
|                                    | High Float Volts (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 55.6V                | _____ V     | ✓ / ✗ |
|                                    | Low Float Volts (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 52.8V                | _____ V     | ✓ / ✗ |
|                                    | Low Load Volts (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 47.0V                | _____ V     | ✓ / ✗ |
|                                    | <p>Depending on the test load available, it may be necessary to adjust the High Load Current alarm threshold down to suit. For example, with 40A test load, adjust the High Load Current threshold (use SM3x Config. software, click on the "Alarms/(Levels)") to 35A. Then simply apply the 40A load and observe the alarm change state.</p> <p>Once the test is complete, be sure to rest the High Load Current to its previous value (or check with the customer for the correct value they require).</p>                                                                                                                                           |                      |             |       |
|                                    | High Load Current (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      | _____ A     | ✓ / ✗ |
|                                    | <p>Temperature alarm tests are performed by heating up (using a heat gun or other source) and cooling down (using a aerosol can of freeze, or a tub of ice) the temperature sensors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |             |       |
|                                    | Battery Temperature High (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      | _____ °C    | ✓ / ✗ |
|                                    | Battery Temperature Low (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      | _____ °C    | ✓ / ✗ |
|                                    | Room Temperature High (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                      | _____ °C    | ✓ / ✗ |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                    |       |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------|-------|
|                             | Room Temperature Low (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  | _____ °C           | ✓ / ✕ |
|                             | <p>When an AC Monitoring PCB is not fitted at system level (as in most cases), the AC Fail alarm is generated from the rectifiers. The rectifiers sense if AC is present, and extend an AC fail alarm to the SM3x. Therefore, to test this alarm, simply turn off the rectifier AC breakers. To allow the monitor to continue to read alarms there must be DC present on the output of the system.</p> <p>As this causes the rectifier output to cease, a Rectifier Fail alarm is also generated. To generate the Urgent Rectifier Fail, turn off the required number to make this occur (usually set to 2, but check via the SM3x Config software for the setting (under the "Rectifier Control" section for the "Control" tab)).</p>                                                                                                                                                                                                                                                                                                                                                                   |  |                    |       |
|                             | AC Fail (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | Urgent             | ✓ / ✕ |
|                             | Rectifier Fail (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  | Non-urgent         | ✓ / ✕ |
|                             | Urgent Rectifier Fail (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | No. Modules: _____ | ✓ / ✕ |
|                             | <p>To check Load MCB fail, connect a load, but with no load turned on. Then switch the breaker to it's off position, and turn on some load (any amount will do). This will cause the alarm to occur as the load side to the circuit will be taken to system common voltage. Turn off the load, and then return the breaker to its on position.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                    |       |
|                             | Load MCB Fail (urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                    | ✓ / ✕ |
|                             | <p>A Rectifier Off Normal alarm is generated with the rectifier enters a state that may degrade its performance (e.g., a single fan fail), or a state that is outside of normal operating conditions (e.g., high temperature, or current limit).</p> <p>You can observe an Off Normal alarm when the system is placed in current limit, or you may wish to stop a fan to generate the alarm. The best way to stop a fan is to use a small cable tie (non-metallic), and push it approximately 1cm through the rectifier grille.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                    |       |
|                             | Rectifier Off Normal (e.g., fan failed) (non-urgent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                    | ✓ / ✕ |
| <b>Tests with Batteries</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                    |       |
|                             | <ul style="list-style-type: none"> <li>• Turn off Battery Breaker/s</li> <li>• Connect battery/batteries</li> <li>• Check the correct Battery Capacity (Ahrs) has been entered (SM3x Config., "Charge" tab, top right of screen). This is the total capacity, so for example 100Ahr strings in parallel, this should be 200.</li> <li>• Go to SM3x Config., "Control" tab. Set the Battery Charge Current Limit to "Enabled" (middle right of screen). Also check Battery Current Limit (BCL) is set to desired level (usually 0.1C<sub>10</sub>, (10%)). This means that for a single 100Ahr battery, the BCL will be 10A, or if two 100Ahr batteries are connected in parallel, the BCL will be 20A.</li> <li>• Go to SM3x Config., "Battery Monitoring" tab. "Tick" Logging Enabled. Set: <ul style="list-style-type: none"> <li>- Status Log Sampling Interval to 7 days (provides a "snapshot" of the battery state every 7 days)</li> <li>- Discharge Log Sampling Interval to 1 minute</li> <li>- Discharge Log Continuation Time to 5 minutes (this is the length of time</li> </ul> </li> </ul> |  |                    |       |

|  |                                                                                                                                                                                                                                                                                                                                                                                              |              |                              |        |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------|--------|
|  | logging continues for after AC power is returned) <ul style="list-style-type: none"> <li>• Connect load (but turn off).</li> <li>• Check <math>V_f</math> is set to 54.0V.</li> <li>• Turn on Battery Breaker/s</li> </ul>                                                                                                                                                                   |              |                              |        |
|  | Check Battery current is positive if charging                                                                                                                                                                                                                                                                                                                                                |              |                              | ✓ / ✗  |
|  | For systems fitted with Battery Monitoring (BCM) cards:<br><br>Go to SM3x Config., "Battery Monitoring" tab and "tick" start scanning.<br>Check that the all the battery cell voltages are approximately the same, at about 2.25V (or 13.5V for 12V monoblocs).<br>If any are out by a large amount, then check the BCM sense wiring is correct.                                             |              |                              | ✓ / ✗  |
|  | Set load to required level (e.g., 30A), Initiate Battery Discharge Test (from SM3x Config., "Charge" tab, click "Battery Test Enabled")                                                                                                                                                                                                                                                      |              | Check operates               | ✓ / ✗  |
|  | Check Battery Current is negative (discharging)                                                                                                                                                                                                                                                                                                                                              |              |                              | ✓ / ✗  |
|  | Check Battery Current                                                                                                                                                                                                                                                                                                                                                                        | Meter: _____ | _____ A                      | ✓ / ✗  |
|  | After several minutes, or until the battery voltage has dropped below approx. 47V, "un-tick" (stop) Battery Test (from SM3x Config., "Charge" tab). During this time the Battery Condition Monitor if fitted) will be logging every minute.                                                                                                                                                  |              |                              |        |
|  | Battery Current Limit (BCL)<br><br>Check that the battery recharge current is limited to the Battery Current Limit level (usually between $0.1C_{10}$ , (10%))<br><br>Note: as the BCL is based on fine voltage control of the system bus, the BCL make take one or two minutes to "settle", i.e., you may observe a brief excursion of the battery recharge current beyond the BCL setting. |              | BCL functions<br><br>_____ % | Yes/No |
|  | Check Manual Equalise (if configured)                                                                                                                                                                                                                                                                                                                                                        |              |                              | ✓ / ✗  |
|  | A Battery MCB Fail alarm is generated from the voltage measured across the battery MCB.<br><br>Therefore, to check a Battery MCB Fail alarm, simply open one of the battery MCB's. If batteries are connected to the system at this time, the alarm may take a few moments to activate. Once test is complete, turn breaker back on.                                                         |              |                              |        |
|  | Battery MCB Fail (urgent)                                                                                                                                                                                                                                                                                                                                                                    |              |                              | ✓ / ✗  |

|                                   |                                                                                                                                                                                                                                                                                      |  |                                                   |        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------|--------|
| <b>Temperature Compensation</b>   |                                                                                                                                                                                                                                                                                      |  |                                                   |        |
|                                   | Ensure Temperature Compensation Enabled                                                                                                                                                                                                                                              |  |                                                   | Yes/No |
|                                   | Apply heat or cold to the Battery Temperature Sensor/s. Check the float voltage moves up or down as expected.                                                                                                                                                                        |  |                                                   | Yes/No |
|                                   | If actual measurement is required, apply a known heat or cold to the sensor. Allow it to fully come to temperature, and record the amount of voltage movement.                                                                                                                       |  | Temp. _____ °C<br><br>Voltage offset: _____ V OK? | Yes/No |
| <b>Check Alarm Relay Contacts</b> |                                                                                                                                                                                                                                                                                      |  |                                                   |        |
|                                   | To generate these alarms, refer to the procedures described earlier in the Commissioning Check-list.<br>Spare relays will not be able to be tested unless an alarm is mapped to them. As these are tested in the factory, it is not essential to test them at time of commissioning. |  |                                                   |        |
| On main SM3x PCB                  |                                                                                                                                                                                                                                                                                      |  |                                                   |        |
|                                   | Relay 1 (Monitor Fail)<br>(pull out the RJ45 lead connecting the monitor to the system rectifier shelf – this simply de-powers the monitor)                                                                                                                                          |  |                                                   | ✓ / ✗  |
|                                   | Relay 2 (Summary Non-urgent)                                                                                                                                                                                                                                                         |  |                                                   | ✓ / ✗  |
|                                   | Relay 3 (Summary Urgent)                                                                                                                                                                                                                                                             |  |                                                   | ✓ / ✗  |
|                                   | Relay 4 (Spare)                                                                                                                                                                                                                                                                      |  |                                                   | ✓ / ✗  |
|                                   | Relay 5 (Spare)                                                                                                                                                                                                                                                                      |  |                                                   | ✓ / ✗  |
|                                   | Relay 6 (Spare)                                                                                                                                                                                                                                                                      |  |                                                   | ✓ / ✗  |

**Note:** Prior to leaving the system after it has been commissioned, ensure all AC, DC and battery circuits are off. If it is required that the system is to be left on (to power load equipment, ensure rectifiers are left in their powered up state, and batteries are in circuit. This will prevent anyone leaving the batteries only powering the load (in which case the batteries would go flat).

## 9.0 MAINTENANCE

As ENATEL Power Systems are state of the art electronic systems, very little routine maintenance is required

### System

- Check all load and battery & alarm cable connections are tight.

### Monitor

- The monitor can give a good indication of the condition of the system. Alarm logs can show issues with the system and rectifiers and should be regularly checked.
- As a minimum, check that the float voltage and load current is as expected.
- If the batteries are fully charged, check the battery current is zero or near to zero amps, and also check that the amp-hours remaining (found in the SM32 Configuration Software application, under the heading "Charge – Battery Capacity") is 100%.

### Rectifiers

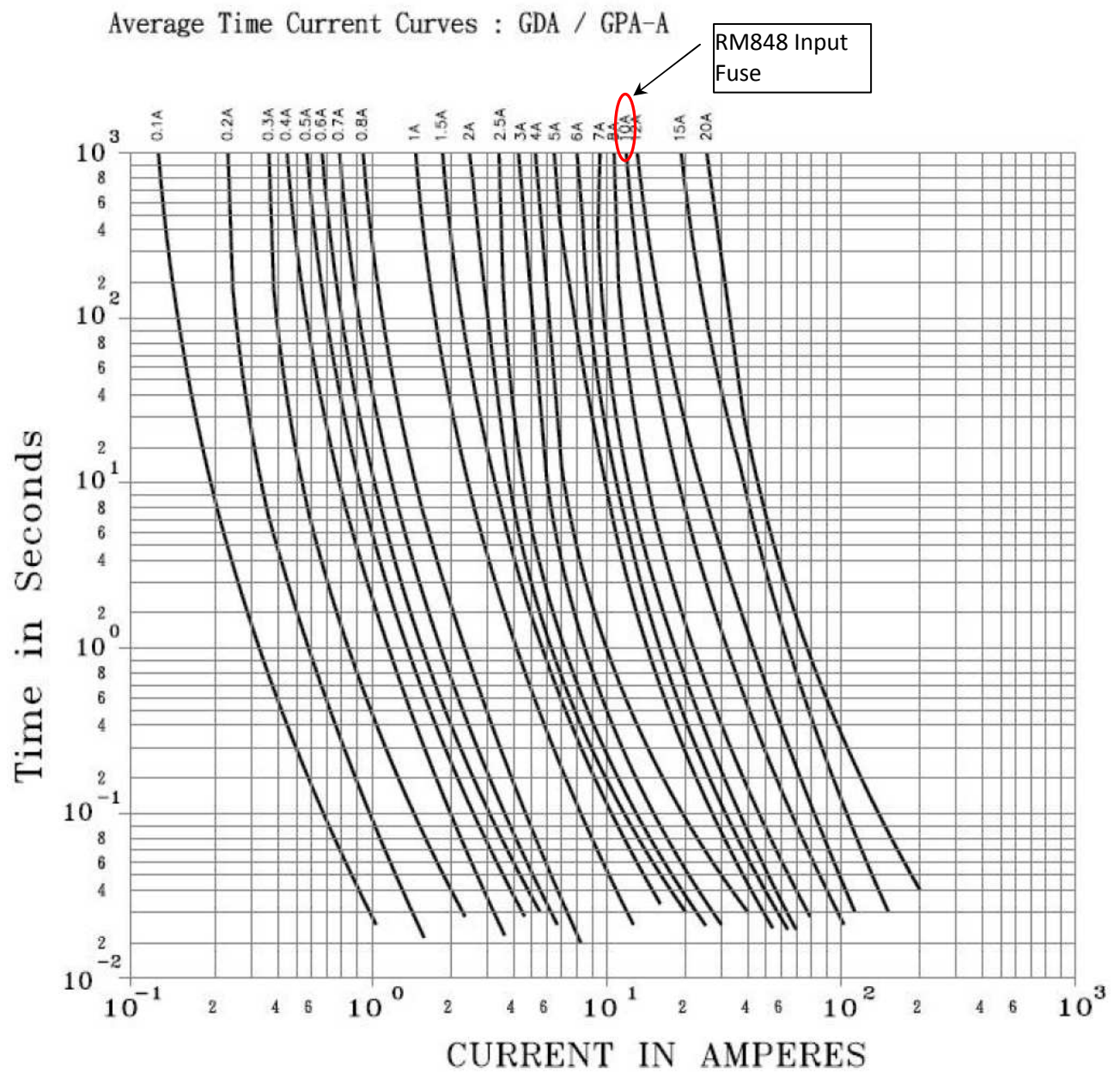
- During normal operation some dust will build-up on the front of the rectifiers. This should be kept to a minimum by regularly wiping the rectifiers to avoid accumulation within the rectifiers and blocking the airflow to the units. The positioning of the system and surroundings will determine the regularity of this requirement.
- In extremely dusty positions it is recommended that the units are removed and cleaned with compressed air to prevent airflow blockages.

### Batteries

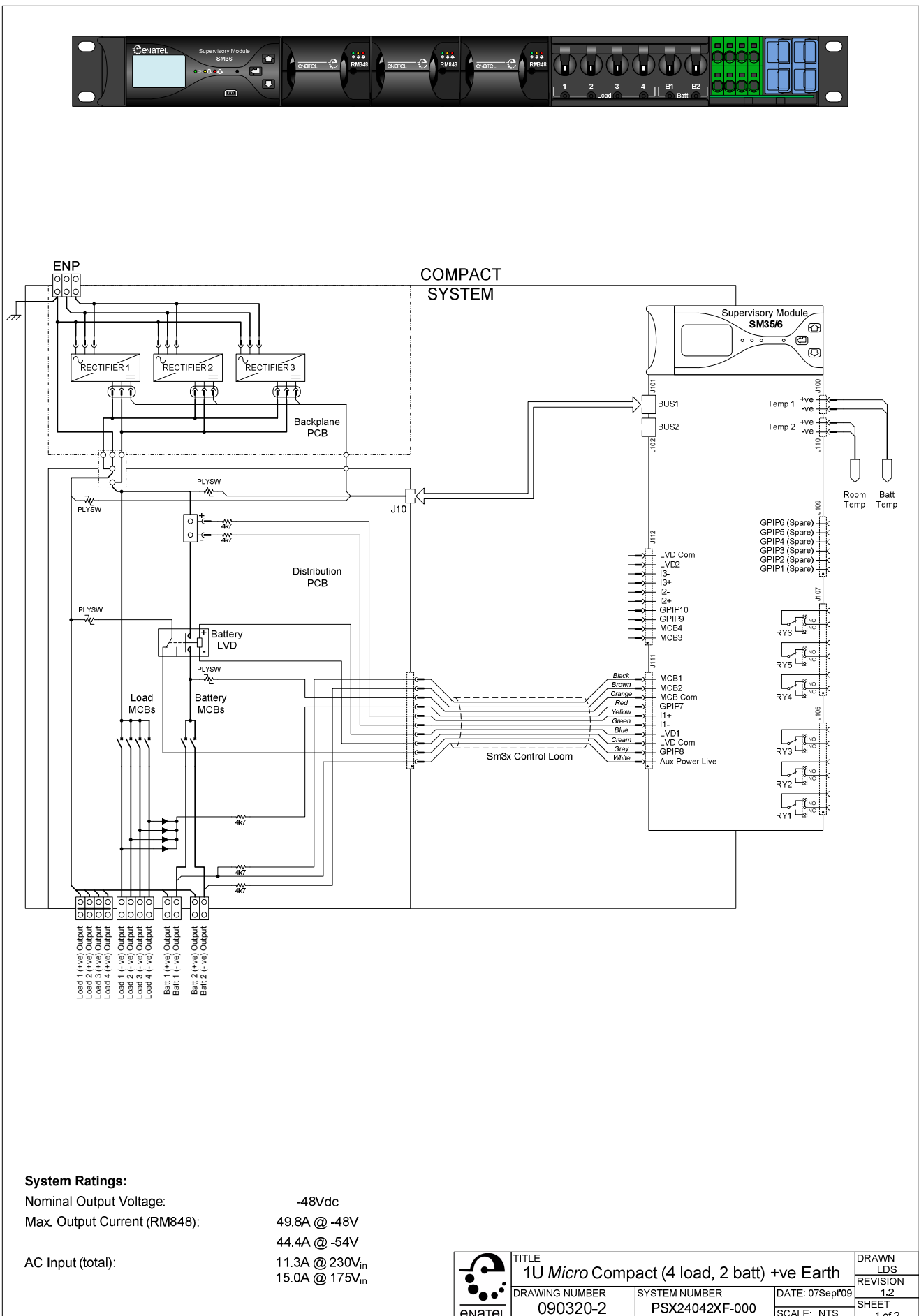
- Battery maintenance will depend on the individual manufacturer's specification, please contact the battery supplier for recommendations.

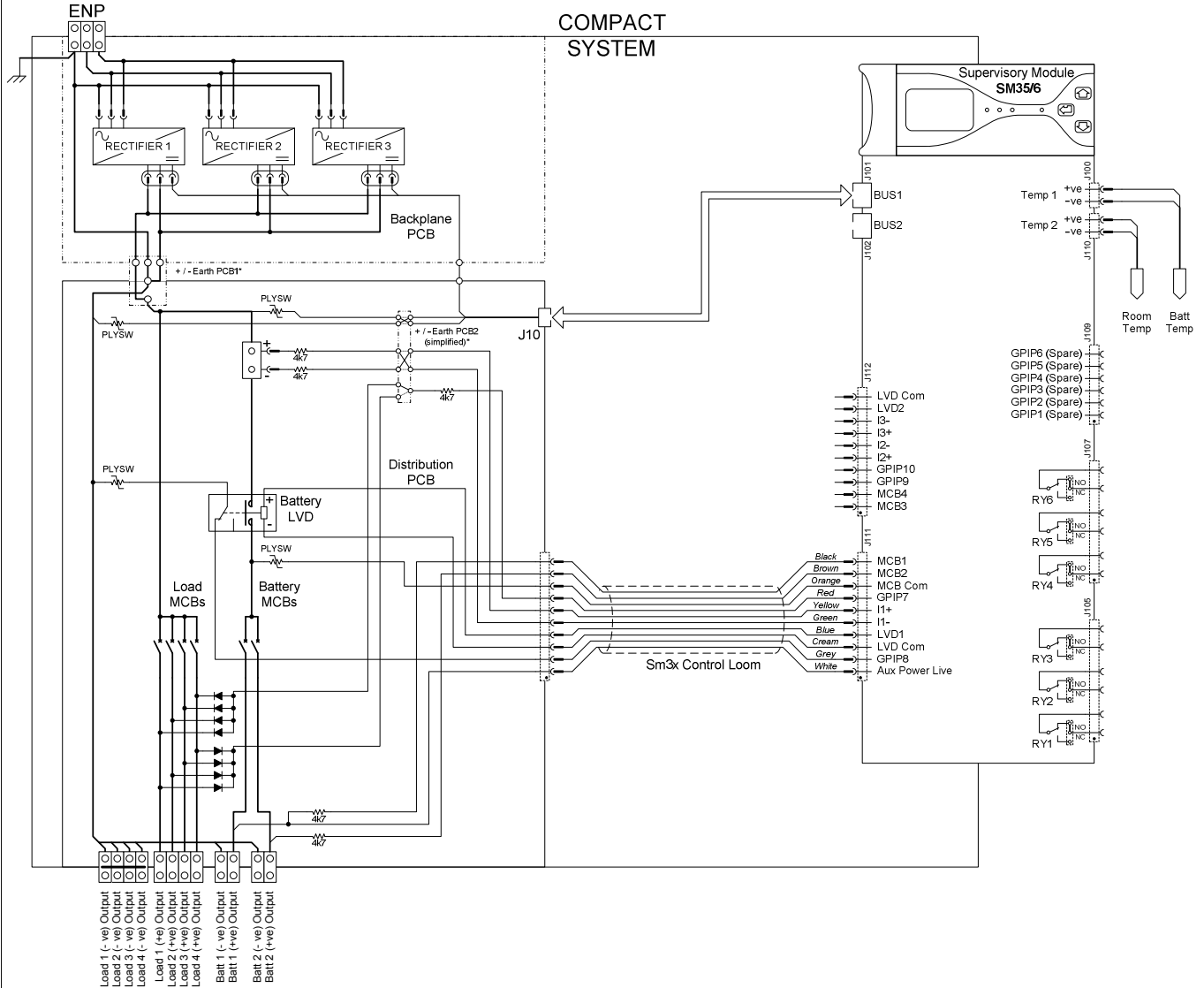
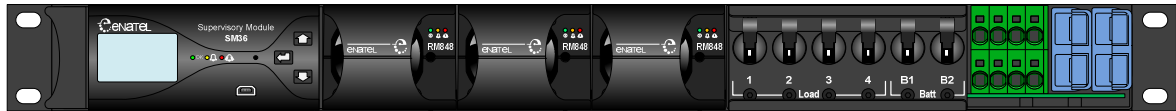
## 10.0 APPENDIX 1 - RECTIFIER INPUT FUSE CURVES

\* The curves are average value, for reference only \*



## 11.0 APPENDIX 2 - SYSTEM WIRING DIAGRAMS





#### System Ratings:

|                              |                            |
|------------------------------|----------------------------|
| Nominal Output Voltage:      | -48Vdc                     |
| Max. Output Current (RM848): | 49.8A @ -48V               |
|                              | 44.4A @ -54V               |
| AC Input (total):            | 11.3A @ 230V <sub>in</sub> |
|                              | 15.0A @ 175V <sub>in</sub> |

\* + / - Earth PCB's 1 & 2 are mechanically interlocked and cannot be placed without being in the correct orientation.  
For more detail please refer to the System manual for more detail

|  |                                                             |                |                |                 |
|--|-------------------------------------------------------------|----------------|----------------|-----------------|
|  | TITLE<br><b>1U Micro Compact (4 load, 2 batt) -ve Earth</b> |                |                | DRAWN<br>LDS    |
|  | DRAWING NUMBER                                              | SYSTEM NUMBER  | DATE: 07Sept09 |                 |
|  | 090320-3                                                    | PSX24042XF-200 | SCALE: NTS     |                 |
|  |                                                             |                |                |                 |
|  |                                                             |                |                | REVISION<br>1.2 |
|  |                                                             |                |                | SHEET<br>1 of 2 |

## 12.0 APPENDIX 3 - MODEL SPECIFICATIONS

### 12.1 System part numbers - 3 Rectifier



| Part number     | Description                    | Load MCB's | Battery MCB's | Voltage VDC | DC Earth | Monitor |
|-----------------|--------------------------------|------------|---------------|-------------|----------|---------|
| PSX2404211F-000 | 1U <i>MICRO</i> Compact System | 4*         | 2*            | 48          | +ve      | SM35    |
| PSX2404211F-200 | 1U <i>MICRO</i> Compact System | 4*         | 2*            | 48          | ±ve      | SM35    |
| PSX2404221F-000 | 1U <i>MICRO</i> Compact System | 4*         | 2*            | 48          | +ve      | SM36    |
| PSX2404221F-200 | 1U <i>MICRO</i> Compact System | 4*         | 2*            | 48          | ±ve      | SM36    |

**\*Note:** Part numbers specify the maximum number of Battery MCB's that can be fitted.

## 12.2 System Naming Convention

### Naming Convention

#### Product type

PSX - Micro Compact System

System Power Rating x 0.1kW

24 - 3x (max) RM848 (3 position shelf)

Number of load MCB's

00 - 0x Load MCB

01 - 1x Load MCB

02 - 2x Load MCB's

03 - 3x Load MCB's

04 - 4x (max\*) Load MCB's (\*maximum for 3 position shelf)

Number of battery

1 - 1x Battery MCB

2 - 2x Battery MCB's

SM3X Type

1 - SM35

2 - SM36

Shelf Height (U)

1 - 1U

Cable Access

F - Front Access

Specific Model Number

-000 - Standard System (positive earth)

-200 - Universal Earth System (negative earth as default)

PSX 24 04 211 F -000